Ziqi Xu

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Research Interests

AI for Healthcare, with a focus on **trustworthy multimodal AI** for personalized prediction and clinical decision support. My research integrates wearable and mobile sensing, clinical text, and EHR data to develop uncertainty-aware, interpretable models for personalized healthcare.

Education

Washington University in St. Louis

Ph.D. in Computer Science (AI/ML) — *May 2027 (expected)*
Advisor: Prof. Chenyang Lu, AI for Health Institute — GPA 4.00/4.00
Dissertation Proposal: *Trustworthy Multimodal AI for Personalized Healthcare*

B.S. in Computer Science — *May 2022*
Second Major: Mathematics — GPA 3.99/4.00

Research Experience

Research Assistant, Washington University in St. Louis — 2022 – Present

Advisor: Prof. Chenyang Lu, AI for Health Institute

- **Trustworthy multimodal AI.** Developed uncertainty-aware learning methods that integrate heterogeneous wearable sensing, ecological momentary assessment (EMA), and clinical data to predict postoperative pain and recovery. Investigated multimodal evidence fusion, uncertainty quantification, and selective prediction to identify when predictions may be unreliable for clinical decision support. Published in *[1]*, *[2]*, and *[3]*.
- **Clinical AI and language models.** Developed machine-learning and language-model-based approaches for risk prediction from structured and unstructured clinical data, with a focus on uncertainty quantification, confidence calibration, and risk-aware prediction to improve the reliability of clinical AI. Published in *[17]* and *[18]*.
- **Interpretable AI for neurosurgical care.** Developed data-driven and clinician-informed methods for patient phenotyping and surgical outcome prediction in neurosurgical populations, including Chiari malformation, syringomyelia, and spine surgery. Examined clinically meaningful patient heterogeneity and interpretable predictors to support individualized surgical decision-making. Published in *[4]*; related work under review in *[U1]*.
- **Translational AI for health.** Collaborated with researchers in Anesthesiology, Neurosurgery, Pain Medicine, and Public Health to formulate clinically motivated machine-learning problems and evaluate models with attention to clinical relevance, interpretability, and potential real-world use.

Applied AI Agents for Hardware Engineering Intern, Apple — May 2026 – Aug 2026

- **Reliable agentic AI systems.** Architected agentic AI systems for complex hardware engineering workflows, integrating graph-based orchestration, tool use, multimodal retrieval, episodic and procedural memory, and simulation control across heterogeneous computing environments to support reliable execution.
- **AI agent evaluation and reliability.** Developed a systematic evaluation framework for knowledge retrieval and grounding across 130 domain-specific questions and five grounding strategies, achieving 96% retrieval success. Incorporated confidence-based evaluation and failure flagging to identify unreliable responses for human review.

Publications

† denotes equal contribution or co-first authorship; ‡ denotes co-senior authorship.

Journal Articles

[1] [Neurosurgery 2026] S. Yakdan†, **Z. Xu**†, J. Zhang, F. Arkam, T. L. Rodebaugh, B. R. Goodin, B. J. Neuman, M. P. Steinmetz, M. P. Kelly, S. Bess, J. Yoon, W. Z. Ray, C. Lu, M. R. Frumkin, J. K. Greenberg. "Real-Time Symptom and Wearable Monitoring May Improve Prediction of Postoperative Recovery in Lumbar Spine Surgery Beyond Standard Measures." *Neurosurgery*, 2026. doi:10.1227/NEU.0000000000004195

[2] [IMWUT/UbiComp 2025] **Z. Xu**, J. Zhang, S. Haroutounian, H. Liu, Z. Cao, G. R. Messner, H. B. Alaverdyan, S. Ahuja, R. Koshy, J. Hanns, M. Frumkin, T. L. Rodebaugh, C. Lu. "Incorporating Uncertainty in Predictive Models Using Mobile Sensing and Clinical Data: A Case Study on Persistent Post-Surgical Pain." *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT / UbiComp)*, vol. 9, no. 2, 2025. doi:10.1145/3729488

[3] [IMWUT/UbiComp 2024] **Z. Xu**, J. Zhang, J. K. Greenberg, M. Frumkin, S. Javeed, J. Zhang, B. Benedict, K. Botterbush, T. L. Rodebaugh, W. Z. Ray, C. Lu. "Predicting Multi-Dimensional Surgical Outcomes with Multi-Modal

Mobile Sensing." *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT / UbiComp)*, vol. 8, no. 2, 2024. doi:10.1145/3659628

[4] [Neurosurgery 2024] V. P. Gupta†, **Z. Xu**†, J. K. Greenberg, J. M. Strahle, G. Haller, T. Meehan, A. Roberts, D. D. Limbrick Jr., C. Lu. "Using Artificial Intelligence to Identify Three Presenting Phenotypes of Chiari Type-I Malformation and Syringomyelia." *Neurosurgery*, 2024. doi:10.1227/neu.0000000000003249

[5] [Neurosurgery 2024] J. K. Greenberg†, M. R. Frumkin†, **Z. Xu**†, J. Zhang, S. Javeed, J. Zhang, B. Benedict, K. Botterbush, S. Yakdan, C. A. Molina, B. H. Pennicooke, D. Hafez, J. I. Ogunlade, N. Pallotta, M. C. Gupta, J. M. Buchowski, B. Neuman, M. Steinmetz, Z. Ghogawala, M. P. Kelly, B. R. Goodin, J. F. Piccirillo, T. L. Rodebaugh, C. Lu, W. Z. Ray. "Preoperative Mobile Health Data Improve Predictions of Recovery from Lumbar Spine Surgery." *Neurosurgery*, vol. 5, no. 3, pp. 548–555, 2024. doi:10.1227/neu.0000000000002911

[6] [Nat. Commun. 2025] C. Najjuuko, **Z. Xu**, S. Kizito, C. Lu, F. M. Ssewamala. "Patterns of Maternal and Child Health Services Utilization and Associated Socioeconomic Disparities in Sub-Saharan Africa." *Nature Communications*, vol. 16, 7840, 2025. doi:10.1038/s41467-025-61350-8

[7] [ACM TCH 2026] J. Zhang, R. Wang, **Z. Xu**, H. Liu, J. Rodriguez, H. Cos, R. Srivastava, L. Raper, D. Sanford, C. Hammill, C. Lu. "Addressing Cohort Variability with Adaptive Fusion of Wearable and Clinical Data: A Case Study in Predicting Pancreatic Surgery Outcomes." *ACM Transactions on Computing for Healthcare*, vol. 7, no. 2, pp. 1–27, 2026. doi:10.1145/3788681

[8] [BMJ Open 2026] K. J. Holzer, H. Alaverdyan, **Z. Xu**, M. R. Frumkin, K. A. Frey, S. H. Gregory, T. L. Rodebaugh, C. Lu, C. R. King, D. Head. "Protocol for Personalised Prediction of Persistent Postsurgical Pain." *BMJ Open*, vol. 16, no. 2, e107055, 2026. doi:10.1136/bmjopen-2025-107055

[9] [AIDS 2025] C. Najjuuko, R. Brathwaite, **Z. Xu**, S. Kizito, C. Lu, F. M. Ssewamala. "Using Machine Learning to Predict Poor Adherence to Antiretroviral Therapy Among Adolescents Living with HIV in Low-Resource Settings." *AIDS*, vol. 39, no. 9, pp. 1204–1213, 2025. doi:10.1097/QAD.0000000000004163

[10] [J. Pain 2025] M. R. Frumkin, J. Zhang, **Z. Xu**, S. Yakdan, B. Benedict, S. Javeed, J. Zhang, K. Botterbush, B. R. Goodin, C. Lu, W. Z. Ray, J. K. Greenberg. "Rethinking Measurement of Movement-Evoked Pain with Digital Technology." *The Journal of Pain*, 105643, 2025. doi:10.1016/j.jpain.2025.105643

[11] [Artif. Organs 2023] N. Shah, B. Xue, **Z. Xu**, H. Yang, E. Marwali, H. Dalton, P. P. R. Payne, C. Lu, A. S. Said, ISARIC Clinical Characterisation Group. "Validation of Extracorporeal Membrane Oxygenation Mortality Prediction & Severity of Illness Scores in an International COVID-19 Cohort." *Artificial Organs*, vol. 47, pp. 1490–1502, 2023.

[12] [Br. J. Anaesth.] M. R. Frumkin, G. R. Messner, K. J. Holzer, **Z. Xu**, T. L. Rodebaugh, H. Bernstein, K. Frey, C. Lu, S. Haroutounian. "Remote Perioperative Symptom Monitoring via Smartphone is Feasible: Evidence from the Personalized Prediction of Persistent Postsurgical Pain (P5) Study." *British Journal of Anaesthesia*, accepted, 2026.

[13] [Spine J. 2025] S. Yakdan, J. Zhang, B. Benedict, **Z. Xu**, S. Javeed, J. K. Zhang, B. A. Steel, V. P. Gupta, K. Botterbush, J. F. Piccirillo, T. L. Rodebaugh, B. R. Goodin, J. M. Buchowski, B. Neuman, D. Hafez, M. P. Kelly, W. Z. Ray, C. Lu, M. R. Frumkin, J. K. Greenberg. "Multi-Domain Postoperative Recovery Trajectories After Lumbar and Thoracolumbar Spine Surgery." *The Spine Journal*, 2025. doi:10.1016/j.spinee.2025.05.017

[14] [Spine Deformity 2025] R. Bhan, S. Yakdan, C. Diaz, J. Zhang, **Z. Xu**, K. Joseph, B. Plog, A. T. Yahanda, J. K. Zhang, S. Javeed, W. Z. Ray, M. Gupta, C. Lu, M. P. Kelly, J. K. Greenberg, B. Neuman. "Association Between Preoperative Activity Levels and Postoperative Complications in Adult Spinal Deformity Surgery." *Spine Deformity*, 2025. doi:10.1007/s43390-025-01158-0

[15] [Clin. Spine Surg. 2025] S. Yakdan, J. Zhang, K. Botterbush, M. I. Kaleem, B. Benedict, **Z. Xu**, M. R. Frumkin, S. Javeed, J. K. Zhang, J. Ogunlade, C. Lu, B. Neuman, M. P. Steinmetz, W. Z. Ray, M. Gupta, J. K. Greenberg. "Sleep Physiology in Spine Disease: A Comparative Study of Cervical and Lumbar Patients." *Clinical Spine Surgery*, 2025. doi:10.1097/BSD.0000000000001926

[16] [JNS Spine 2023] J. K. Greenberg, S. Javeed, J. K. Zhang, B. Benedict, M. R. Frumkin, **Z. Xu**, J. Zhang, T. L. Rodebaugh, C. Lu, M. Steinmetz, Z. Ghogawala, M. Bydon, W. Z. Ray. "Current and Future Applications of Mobile Health Technology for Evaluating Spine Surgery Patients: A Review." *Journal of Neurosurgery: Spine*, 2023. doi:10.3171/2022.12.SPINE221302

Conference Proceedings

[17] [ACL Main 2026] S. Wang, **Z. Xu**, C. Najjuuko, C. Alba, C. Lu. "CURA: Clinical Uncertainty Risk Alignment for Language Model–Based Risk Prediction." *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. doi:10.18653/v1/2026.acl-long.1567

[18] [ACL Findings 2025] B. C. Warner, **Z. Xu**, S. Haroutounian, T. Kannampallil, C. Lu. "Utilizing Semantic Textual Similarity for Clinical Survey Data Feature Selection." *Findings of the Association for Computational Linguistics (ACL)*, pp. 502–520, 2025. doi:10.18653/v1/2025.findings-acl.27

[19] [KDD 2023] B. Xue, A. Said, **Z. Xu**, H. Liu, N. Shah, H. Yang, P. Payne, C. Lu. "Assisting Clinical Decisions for Scarcely Available Treatment via Disentangled Latent Representation." *ACM SIGKDD Conference on Knowledge Discovery & Data Mining (KDD)*, 2023. doi:10.1145/3580305.3599774

Manuscripts Under Review

[U1] [Child's Nervous System] **Z. Xu**, W. Mualem, S. Venigalla, J. K. Greenberg, T. Meehan, J. M. Strahle, J. L. Roland, G. Haller, D. D. Limbrick Jr.‡, C. Lu‡. "Expert Opinion or Data-Driven Features? AI-Based Prediction of Headache Outcomes from Surgery for Chiari 1 Malformation with Syringomyelia." *Child's Nervous System*, under review, 2026. *Corresponding author.*

[U2] [J. Alzheimer's Dis.] K. Patel†, **Z. Xu**†, M. Dolatshahi, P. Gao, L. Lloyd, P. K. Commean, C. Lu, C. A. Raji. "Machine Learning Prediction of Amyloid Accumulation Linked to Adiposity for Alzheimer's Disease Assessment." *Journal of Alzheimer's Disease*, under major revision, 2026.

[U3] [JAMIA] C. Najjuuko, L. Jin, **Z. Xu**, M. Yu, J. Siegler, M. Middleton, R. Stoler, C. Lu. "Uncertainty-Aware AI Decision Support for Emergency Medical Services Using Conformal Prediction." *Journal of the American Medical Informatics Association (JAMIA)*, under review, 2026.

Conference Abstracts & Review Articles

[A1] [Neurosurgery 2026] S. Yakdan†, **Z. Xu**†, J. Zhang, J. F. Piccirillo, T. L. Rodebaugh, G. Burel, B. J. Neuman, M. P. Steinmetz, W. Z. Ray, C. Lu. "Leveraging Preoperative Mobile Health Data to Predict 1-Year Recovery Outcomes After Lumbar Spine Surgery." *Neurosurgery*, vol. 72, suppl. 1, p. 43, 2026. doi:10.1227/neu.0000000000003964_202

[A2] [Neurosurgery 2025] J. Zhang, S. Yakdan, B. C. Benedict, J. Zhang, M. I. Kaleem, S. Javeed, K. S. Botterbush, M. R. Frumkin, **Z. Xu**, C. Lu. "Minimum Clinically Important Differences in Quantitative Objective Markers of Physical Function in Patients Undergoing Lumbar Spine Surgery." *Neurosurgery*, vol. 71, suppl. 1, pp. 99–100, 2025. doi:10.1227/neu.0000000000003360_414

[A3] [Semin. Spine Surg. 2023] S. Yakdan, K. Botterbush, **Z. Xu**, C. Lu, W. Z. Ray, J. K. Greenberg. "Machine Learning and Lumbar Spondylolisthesis." *Seminars in Spine Surgery*, vol. 35, no. 3, 101048, 2023. doi:10.1016/j.semss.2023.101048

Invited Talks and Presentations

Invited Talks

"Beyond Prediction: Trustworthy Multimodal AI for Pain." 2026
#Pain2026 AI and Pain Workshop, Gulf Coast Consortia

"Using AI to Predict Post-Surgical Headache Improvement for Chiari I and Syringomyelia." 2026
Park-Reeves Chiari & Syringomyelia Research Center Annual Update

"Expert Opinion or Data-Driven Features? AI-Based Prediction of Headache Outcomes from Surgery for Chiari 1 Malformation with Syringomyelia." 2026
Bobby Jones Chiari & Syringomyelia Foundation Think Tank

Conference Presentations

"Trustworthy Agents for Clinical Workflows: Uncertainty-Guided Delegation in a Heterogeneous Collaborative System." 2026
WashU AI for Health Symposium, St. Louis, MO. (Poster Presentation)

"Incorporating Uncertainty in Predictive Models Using Mobile Sensing and Clinical Data: A Case Study on Persistent Post-Surgical Pain." 2025
ACM UbiComp/IMWUT, Helsinki, Finland. *[2]*

"Predicting Multi-Dimensional Surgical Outcomes with Multi-Modal Mobile Sensing." 2024
ACM UbiComp/IMWUT, Melbourne, Australia. *[3]*

Honors and Awards

WashU Big Ideas Competition – $50K Research Award (team member) 2025

Doctoral Honors – Top 15–20% in WashU's Periodic Review of Doctoral Students 2024–2026

Outstanding Reviewer – *Journal of Digital Health* 2025

Dean's Select Ph.D. Fellowship – Washington University in St. Louis 2022–2027

Summa Cum Laude – Washington University in St. Louis 2022
Dean's List – WashU School of Engineering 2020–2022

Grant and Proposal Experience

Research Awards and Supported Proposals

When Should Clinical AI Agents Act? Uncertainty-Aware Delegation for Reliable Decision-Making, Anthropic AI for Science Program; $5,000 in Research Credits 2026-2027
Proposal lead. Selected for research support to investigate uncertainty-aware delegation among heterogeneous AI agents for reliable clinical decision-making in clinical workflows.

Leveraging Machine Learning and Wearable Device Activity Patterns to Improve Prehabilitation and Surgical Outcomes, BJC HealthCare Innovation Lab & WashU I2DB Big Ideas Competition; $50,000; Team Leads: D. Sanford and C. Lu 2025–2026
Team member. Developed wearable-activity prediction models and contributed the corresponding methods and results to the funded proposal.

Contributions to Funded Grant Proposals

Contributed predictive modeling, preliminary analyses, and technical writing to the following funded proposals; not named personnel on these awards.

IMPETUS: Integrated Mechanisms, Phenotypes, and Translational Underpinnings of Chronic Pain after Surgery, NIH NINDS RM1NS135283; MPI: Haroutounian, Creed, Lu, Rodebaugh, Shepherd, Sinha 2024
Developed multimodal machine-learning models for chronic postsurgical pain phenotyping and contributed the associated methods and preliminary results to the proposal.

Redefining Chiari Type I Malformation and Its Impact on Brain Development, NIH NINDS P01NS131131; MPI: D. D. Limbrick Jr. and C. Lu 2023
Developed phenotyping and outcome-prediction models and contributed the corresponding methods and preliminary results to the proposal.

Teaching Experience

Guest Lecturer, *CSE 419A: Introduction to AI for Health* Fall 2025, Fall 2026
Guest Lecturer, *PHS 5100: Development, Validation and Application of Risk Prediction* Spring 2026
Assistant Instructor, *CSE 419A: Introduction to AI for Health* Fall 2024
Assistant Instructor, *CSE 531A: AI for Health* (graduate) Spring 2024
Teaching Assistant, *CSE 417: Introduction to Machine Learning* Spring 2023
Teaching Assistant, *CSE 240: Logic and Discrete Mathematics* Spring 2020

Research Mentoring

Mentored eight students (one Ph.D. student, one Ph.D. rotation student, one M.S. student, four undergraduate students, and one medical student) in multimodal sensing, uncertainty quantification, and clinical machine learning, in collaboration with Prof. Chenyang Lu. **Five mentees co-authored publications, including three as first or co-first authors**; two received institutional research or academic awards.

Ph.D. and Graduate Students

Claire Najjuuko, Ph.D. student; co-advised by Prof. Chenyang Lu and Prof. F. M. Ssewamala 2023–Present
Mentored the machine-learning components of dissertation research on maternal–child health and antiretroviral therapy adherence in Sub-Saharan Africa. **First author** on *[6]* (*Nature Communications*) and *[9]* (*AIDS*); recipient of the 2025 Provost's Research Excellence Award.

Sizhe Wang, Ph.D. rotation student, Computer Science 2025
Mentored research on uncertainty–risk alignment for language-model-based clinical prediction. **First author** on *[17]* (ACL 2026). Now a Ph.D. student at Washington University in St. Louis.

Yuyang Wu, M.S. Computer Science 2025
Mentored research on uncertainty quantification for language-model-based clinical prediction. Master's thesis:

Quantifying Uncertainty in Clinical Language Models: When and How It Is Useful for Clinical Prediction. Now Machine Learning Engineer at TikTok.

Undergraduate and Medical Students

Lucien Liu, B.S. Computer Science 2026
Mentored research on rhythm-based feature extraction from ecological momentary assessment (EMA) time series. Presented at the WashU Undergraduate Research Symposium; manuscript in preparation. Now M.S. student at Massachusetts Institute of Technology.

Kavya Patel, B.S. Computer Science; Minor in Chemistry 2025
Mentored research on machine-learning prediction of amyloid accumulation and its association with adiposity in Alzheimer's disease. **Co-first author** on *[U2]*, under major revision at *Journal of Alzheimer's Disease*. Now Associate Consultant at Bain & Company.

Siddharth Venigalla, M.D. candidate 2025
Mentored contributions to data processing, experimental design, and interpretation for predicting headache outcomes following surgery for Chiari I malformation with syringomyelia. Co-author on *[U1]*, under review.

Tongyang (Jason) Wang, B.S. Computer Science 2025–Present
Mentoring research on wearable-sensing models for predicting adverse postoperative outcomes in the SPAR cohort; manuscript in preparation. Recipient of the 2026 Outstanding Junior Award for Academic Excellence, WashU Department of Computer Science & Engineering.

Zihan Cao, B.S. Computer Science 2023
Mentored research on uncertainty-aware multimodal modeling for persistent postsurgical pain prediction. Co-author on *[2]* (ACM IMWUT/UbiComp 2025). Now M.S. student at Duke University.

Professional Service

Program Committees

- Program Committee, ACM CIKM Workshop on Large-scale AI for Sensing Systems (LASS), 2026.
- Program Committee, International Multi-Conference on Artificial Intelligence Technology, 2026.

Journal and Conference Reviewing

- **Conference reviewing:** ICLR, 2027. ACL Rolling Review, 2026.
- **Journal reviewing:** *NPJ Digital Medicine* (2026); *Nature Communications* (early-career co-reviewer); *ACM Transactions on Computing for Healthcare* (2026); *Scientific Reports* (2024–2026); *BMC Medical Informatics and Decision Making* (2025–2026); *BMC Anesthesiology* (2026); *Journal of Digital Health* (2025).

Departmental and Community Service

- **Graduate Student Advisory Board Member**, WashU Center for Career Engagement, 2026–2027.
- **Volunteer Organizer**, WashU AI for Health Symposium (2024–2026) and AI for Health Venture Day (2026).
- **Test Writer & Tournament Volunteer**, WashU Science Olympiad, 2026–2027. Develop and proctor STEM competition events supporting the WashU Invitational (1,800+ middle- and high-school students) and Missouri Regional Tournament.

Technical Skills

Methods: multimodal learning, uncertainty quantification and calibration, selective prediction, longitudinal and time-series modeling, multitask learning, causal inference, LLM fine-tuning (LoRA), retrieval-augmented generation (RAG), and agentic AI.
Programming & Tools: Python, PyTorch, scikit-learn, R, SQL, LangGraph, Git, and high-performance computing.
Health Data: wearable and mobile health data, ecological momentary assessments (EMA), electronic health records (EHR), clinical notes, and patient-reported outcomes.